ENLARGING A SITE GIVES A SUBTOPOS WITH A RETRACTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The proposition producing a morphism of sites from a functor. Same chapter, Proposition proposition-get-morphism. Shows that the functor defines a morphism of sites.
- The pushforward and pullback computed through a right adjoint. Same chapter, Lemma lemma-have-functor-other-way and Lemma lemma-when-shriek. Supply the formulas for the pushforward and pullback of the morphism of topoi.
- The two composites of the pullback with the pushforward. Same chapter, Lemma lemma-back-and-forth. Shows that the two composites are isomorphic to the identity.

Source reference. Original proposition 00X6: sources/stacks-sites/pinned-original-source.tex, source lines 4978–4978. Navigation pointer invoked at proof line 4978; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XC: sources/stacks-sites/pinned-original-source.tex, source lines 4980–4980. Navigation pointer invoked at proof line 4980; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00XR: sources/stacks-sites/pinned-original-source.tex, source lines 4897–4897. Navigation pointer invoked at proof line 4897; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XT: sources/stacks-sites/pinned-original-source.tex, source lines 4986–4986. Navigation pointer invoked at proof line 4986; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that*

(a) *u is cocontinuous,*
(b) *u is continuous,*
(c) *u is fully faithful,*
(d) *fibre products exist in $\mathcal{C}$ and u commutes with them, and*
(e) *there exist final objects $e_{\mathcal{C}} \in \mathrm{Ob}(\mathcal{C})$, $e_{\mathcal{D}} \in \mathrm{Ob}(\mathcal{D})$ such that $u(e_{\mathcal{C}}) = e_{\mathcal{D}}$.*

Let $g_!, g^{-1}, g_*$ be as above. Then, u defines a morphism of sites $f : \mathcal{D} \to \mathcal{C}$ with $f_* = g^{-1}$, $f^{-1} = g_!$. The composition

$$Sh(\mathcal{C}) \xrightarrow{g} Sh(\mathcal{D}) \xrightarrow{f} Sh(\mathcal{C})$$

is isomorphic to the identity morphism of the topos $Sh(\mathcal{C})$. Moreover, the functor f^{-1} is fully faithful.

Proof. By assumption the functor u satisfies the hypotheses of Proposition 00X6. Hence u defines a morphism of sites and hence a morphism of topoi f as in Lemma 00XC. The formulas $f_* = g^{-1}$ and $f^{-1} = g_!$ are clear from the lemma cited and Lemma 00XR. We have $f_* \circ g_* = g^{-1} \circ g_* \cong \mathrm{id}$, and $g^{-1} \circ f^{-1} = g^{-1} \circ g_! \cong \mathrm{id}$ by Lemma 00XT.

We still have to show that f^{-1} is fully faithful. Let $\mathcal{F}, \mathcal{G} \in \mathrm{Ob}(Sh(\mathcal{C}))$. We have to show that the map

$$\mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Mor}_{Sh(\mathcal{D})}(f^{-1}\mathcal{F}, f^{-1}\mathcal{G})$$

is bijective. But the right hand side is equal to

$$\begin{aligned} \mathrm{Mor}_{Sh(\mathcal{D})}(f^{-1}\mathcal{F}, f^{-1}\mathcal{G}) &= \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}, f_* f^{-1}\mathcal{G}) \\ &= \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}, g^{-1} f^{-1}\mathcal{G}) \\ &= \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{F}, \mathcal{G}) \end{aligned}$$

(the first equality by adjunction) which proves what we want. □